Rational Functions: Canceled-Factor Holes vs Vertical Asymptotes

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The diagnosis, in the order you should run it:

- Factor the numerator and the denominator completely. Do not cancel yet.
- Every zero of the *original* denominator is a break in the domain — the function is undefined there either way.
- A denominator factor that cancels **completely** leaves a *hole*: the height is the reduced form evaluated at that x.
- A denominator factor that **survives** the reduction gives a *vertical asymptote*: the values grow without bound there.
- A repeated factor cancels only as many times as it appears on top. $(x-2)^2$ over one $(x-2)$ still leaves one in the denominator.

For every find-and-fix item, name the factor that was mishandled before you write the corrected answer.

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1. Factor both parts of $f(x) = \dfrac{x^2-9}{x^2+x-12}$ completely. Do not reduce yet — write the two factored polynomials, and circle the factor that appears in both.

Answer: ____________________

2. Reduce $f(x) = \dfrac{x^2-9}{x^2+x-12}$ to lowest terms.
State the restriction on the domain that the reduced form no longer shows.

Answer: ____________________

3. The factor $x - 3$ cancels in $f(x) = \dfrac{x^2 - 9}{x^2 + x - 12}$, so the graph has a hole there.
Find the *height* of that hole, i.e. the value the outputs approach as x approaches 3. Give an exact fraction.

Answer: ____________________

4. Still with $f(x) = \dfrac{x^2 - 9}{x^2 + x - 12}$: one denominator factor survives the reduction. Solve for the x-value of the vertical asymptote.

Answer: ____________________

5. **Find and fix the mistake.** Nina factors $g(x) = \dfrac{x^2 - 25}{x^2 - 3x - 10}$ and writes: "The denominator is zero at $x = 5$ and $x = -2$, so there are vertical asymptotes at both."
Name the factor Nina mishandled. Then give the one real vertical asymptote and the exact coordinates of the hole.

Answer: ____________________

6. Find and fix the mistake. Ray looks at $h(x) = \dfrac{2x^2 - 8}{x^2 - 4x + 4}$, sees a factor of $x - 2$ on the top and on the bottom, and concludes there is a hole at $x = 2$.
Factor the denominator completely, reduce h, and state what is really at $x = 2$.

Answer: ____________

7. Both at once. For $k(x) = \dfrac{x^2 - x - 6}{x^2 - 4}$, find the exact coordinates of the hole and the equation of the vertical asymptote. Say which factor produced each.

Answer: ____________

8. Explain. A classmate says, "If the denominator is zero there, the graph shoots off to infinity — that is what a zero denominator means." Write a short paragraph explaining why that is only half true. Use one-sided behaviour or a table of nearby outputs to contrast the two cases, and say explicitly whether the function has a value at a hole.